

Sep 15 21:24

78%

saad@sand-dweller: ~/xv6-labs-2026 — make qemu

~/xv6-labs-2026

```
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcontrol=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/sixfive.o user/sixfive.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sixfive user/sixfive.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -t user/_sixfive | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/sixfive.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcontrol=medany -ffreestanding -fno-com
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iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/mendump.o user/mendump.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_mendump user/mendump.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -t user/_mendump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/mendump.sym
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iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/find.o user/find.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_find user/find.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -t user/_find > user/find.o
riscv64-linux-gnu-objdump -t user/_find | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/find.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcontrol=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/uptime.o user/uptime.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_uptime user/uptime.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -t user/_uptime > user/uptime.o
riscv64-linux-gnu-objdump -t user/_uptime | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/uptime.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombi
e user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_mendump user/_find user/_uptime user/_mendump
ballocc: First 1220 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -n 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,ide=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
```

xv6 kernel is booting

```
hart 2 starting
hart 1 starting
init: starting sh
$ uptime
uptime: 134 ticks
$ uptime
uptime: 284 ticks
$ uptime
exec uptime failed
$ uptime
uptime: 551 ticks
$ uptime
uptime: 768 ticks
$ uptime
uptime: 799 ticks
$ uptime
uptime: 822 ticks
$ uptime
uptime: 852 ticks
$ uptime
uptime: 1236 ticks
$
```

```
Sep 15 21:39
saad@sand-dweller: ~/xv6-labs-2026
xv6
sleep      2 25 35368
starfive   2 26 37568
memdump    2 27 43608
find        2 28 44560
uptime     2 29 35088
memdump    2 27 43608
console    3 31 0
$ find ..
usage: find path name
$ find . .
./README
./findtest.sh
./sixfive.txt
./cat
./echo
./forktest
./grep
./init
./kill
./ln
./ls
./mkdir
./rm
./sh
./stressfs
./usertests
./grind
./hw
./zombie
./logstress
./forphan
./dorphan
./sync
./sleep
./sixfive
./memdump
./find
./uptime
./memdump
./console
$ find . find
./findtest.sh
./find
$ $ find . ^find$
exec $ failed
$ find . ^find$
exec $ failed
$ find . ^find$
./find
$ QEMU: Terminated
saad@sand-dweller:~/xv6-labs-2026$ git status
On branch assign01
Your branch is up to date with 'origin/assign01'.

Changes not staged for commit:
```

```
Sep 15 22:10
saad@sand-dweller: ~/xv6-labs-2026 — make qemu
~/xv6-labs-2026
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/cat user/echo user/forktest user/grep user/init user/kill user/ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_mendump user/_find user/_uptime user/_mendump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1229 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ ls
.          1 1 1624
..         1 1 1624
README    2 2 2441
findtest.sh 2 3 91
sixfive.txt 2 4 20
cat        2 5 36728
echo       2 6 35592
forktest   2 7 18136
grep       2 8 44184
init       2 9 35976
kill       2 10 35512
ln         2 11 35384
ls         2 12 42880
mkdir      2 13 35660
rm         2 14 35552
sh         2 15 38632
stressfs   2 16 36416
usertests  2 17 205800
grind      2 18 51824
wc         2 19 37728
zombie     2 20 34848
logstress  2 21 37500
forphan    2 22 30352
dorphan    2 23 35800
sync       2 24 34944
sleep      2 25 35360
sixfive    2 26 37560
mendump    2 27 43600
find       2 28 44560
uptime     2 29 35888
mendump    2 27 43600
console    3 31 0
$ sh < findtest.sh
usage: find path name
$ cat findtest.sh
mkdir a
echo hello > a/b
mkdir c
echo hello > c/b
echo hello > b
find . b -exec grep hello
$
```

```
Sep 15 23:17
saad@sand-dweller: ~/xv6-labs-2026 — make qemu
balloos: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ wait
$ wait
$ wait
exec wait failed
$ ls
.          1 1 1624
..         1 1 1624
README    2 2 2441
findtest.sh 2 3 91
sixfive.txt 2 4 20
cat        2 5 36728
echo       2 6 35592
forktest   2 7 18136
grep       2 8 44184
lnit       2 9 35976
kill       2 10 35512
ln         2 11 35384
ls         2 12 42880
mkdir      2 13 35860
rm         2 14 35552
sh         2 15 67472
stressfs   2 16 36416
usertests  2 17 285800
grind      2 18 51824
wc         2 19 37728
zombie     2 20 34848
logstress  2 21 37500
forphan    2 22 30352
dorphan    2 23 35800
sync       2 24 34944
sleep      2 25 35360
sixfive    2 26 37560
memdump    2 27 43600
find       2 28 44560
uptime     2 29 35888
memdump    2 27 43600
console    3 31 0
$ uptime
uptime: 103 ticks
$ history
1 wait
2 wait
3 wait
4 ls
5 uptime
6 history
$
```